Java Programming - Internal Exam Answers

1. What is Java? Explain the architecture of Java...

Java is a high-level, class-based, object-oriented programming language that is platform-independent due to its use of the Java Virtual Machine (JVM).

Java Architecture:

- Source Code (.java) is compiled by javac into Bytecode (.class)
- Bytecode is executed by the Java Virtual Machine (JVM)
- JVM interacts with JRE and JDK

Features/Objectives:

- Simple, Object-Oriented, Platform-Independent, Secure, Robust, Multithreaded, Portable, High Performance, Distributed

OOP Significance:

- Encourages modular, reusable, and scalable code using inheritance, encapsulation, abstraction, and polymorphism.

2. What is inheritance? Explain different types/forms...

Inheritance is a mechanism in OOP where a subclass inherits from a superclass.

Types of Inheritance:

- Single, Multilevel, Hierarchical (Java does not support multiple inheritance with classes)

Example (Single Inheritance):

```
class Animal {
   void sound() { System.out.println("Animal makes a sound"); }
}
class Dog extends Animal {
   void bark() { System.out.println("Dog barks"); }
}
```

Features/Objectives:

- Code Reusability, Method Overriding, Extensibility, Polymorphism Support, Maintainability

3. Explain method overloading and operator overloading...

Method Overloading:

Defining multiple methods with same name but different parameters.

Example:

```
class Demo {
    void show(int a) { System.out.println("int: " + a); }
    void show(String b) { System.out.println("String: " + b); }
}
```

Operator Overloading:

Not supported in Java (except + for Strings).

Difference:

Aspect	Method Overloading	Operator Overloading
Support	Supported	Not supported
Based on	Method name and parameters	Operator and operand types
Purpose	Code readability	Custom behavior for operators